INCH-POUND

MIL-C-52705B
28 November 1994
SUPERSEDING
MIL-C-52705A
26 May 1977

MILITARY SPECIFICATION

COUPLINGS: HOSE, STEAM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers steam hose couplings for conveyance of saturated and superheated steam. The working pressure of the couplings shall be at least 250 psig (pounds per square gauge) at a temperature of 450 °F.

1.2 Classification. The couplings shall be of the following types and sizes, as specified (see 6.2):

Type A - Expansion ring coupling with slotted swivel.
Size - 3/4 (.75) inch.

Type B - Ribbed shank coupling with hexagonal swivel.
Sizes - 1/2 (.50) inch, 3/4 (.75) inch, 1 (1.0) inch, 1-1/4 (1.25) inch and 1-1/2 (1.50) inch.

Type C - Ribbed shank coupling with wing nut swivel.
Sizes - 1/2 (.50) inch, 3/4 (.75) inch, 1 (1.0) inch, 1-1/4 (1.25) inch, 1-1/2 (1.50) inch, 2 (2.0) inch, and 2-1/2 (2.50) inch.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: MOBILITY TECHNOLOGY CENTER - BELVOIR, ATTN AMSTA RBES, 10115 GRIDLEY RD, STE 128, FT BELVOIR VA 22060-5843 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4730

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-H-775 - Hose, Hose Assemblies; Rubber, Plastic, Fabric, or Metal (Including Tubing); and Associated Hardware; Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-130 - Identification Marking of US Military Property.

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the STDZN DCMNT ORDER DESK, BLDG 4D, 700 ROBBINS AVE, PHILADELPHIA PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 47M - Ferritic Mallleable Iron Castings (Metric).
B 16 - Free-Cutting Brass Rod, Bar, and Shapes for Use in Screw Machines.
B 21 - Naval Brass Rod, Bar, and Shapes.
B 124 - Copper and Copper-Alloy Forging Rod, Bar, and Shapes.
B 135 - Seamless Brass Tube.
B 453 - Copper-Zinc-Lead Alloy (Leaded-Brass) Rod.

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B 584	-	Copper Alloy Sand Castings for General Applications.
B 633	-	Zinc on Iron and Steel, Electrodeposited Coatings of.
D 2000	-	Standard Classification System for Rubber Products in Automotive Applications.
E 172	-	Describing and Specifying the Excitation Source in Emission Spectrochemical Analysis.

(Application for copies should be addressed to: AMERCN SCTY & MTRLS, 1916 RACE STRET, PHILADELPHIA PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The couplings shall be as shown in figure 1 through figure 4 and as specified herein.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The steam hose couplings shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the couplings may be exposed.

3.3.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

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3.3.3 Couplings.

3.3.3.1 Castings. Cast parts of couplings shall conform to ASTM B 584, copper alloy UNS no. C85200, C85400, or C85700. Type C coupling components shall be malleable iron conforming to ASTM A 47M, grade optional. Castings with holes are not acceptable.

3.3.3.2 Rods. Copper alloy rods for manufacturing of coupling components shall conform to ASTM B 16; ASTM B 21, alloy optional; ASTM B 124, copper alloy no. C46200, C46400, or C48200; or ASTM B 453, alloy optional.

3.3.4 Expansion rings. Expansion rings shall be made from seamless drawn tubing conforming to ASTM B 135, copper alloy UNS no. C23000, C26000, C27000, C27200, C28000, C33000, C33200, or C37000.

3.3.5 Rubber washers. The rubber washers shall be molded from heat-resistant and water-resistant synthetic rubber stock conforming to ASTM D 2000 M3DA A26B36C32EA14G21 and shall conform to the dimensions shown in figures 2 and 3.

3.3.6 Type B coupling parts and clamps. All type B coupling parts and clamps shall be brass conforming to ASTM B 584, copper alloy no. C85200.

3.4 Finish. Outside surfaces of sandcastings shall be made smooth by shotblasting or sandblasting. The inside surfaces of all parts, except the corrugations in male and female sections, shall be machine finished.

3.4.1 Finish on type C couplings. Unless otherwise specified (see 6.2), type C coupling components, except washers and soft metal contact surface inserts, shall be zinc plated in accordance with the requirements of ASTM B 633, Fe/Zn 25, type II.

3.5 Corrugations. The corrugations in male and female sections shall be left rough so as to bind the hose.

3.6 Threads. Type A and type B couplings shall be threaded NPSH and type C couplings shall be threaded NPT. All threads shall be in accordance with FED-STD-H28.

3.7 Parts. A complete set of couplings of the sizes specified (see figures 2, 3 and 4) shall consist of the following parts:

Type A - Part 7 - Female section.
- Part 8 - Male section.
- Part 3 - Swivel of female section.
- Part 4 - Expansion rings (one each for male and female sections).
- Part 5 - Rubber washer fittings recess in swivel.

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Type B - Part 1 - Female section.
- Part 2 - Male section.
- Part 3 - Swivel of female section.
- Part 4 - Clamps for attaching hose.
- Part 5 - Washer fitting recess in swivel.

Type C - Part 9 - Coupling.

3.8 Hydrostatic pressure. When tested in accordance with 4.5.2.1, the couplings shall be capable of withstanding a hydrostatic pressure of 400 pounds per square inch (psi).

3.9 Identification marking. The couplings shall be identified in accordance with MIL-STD-130.

3.10 Workmanship. Material shall be uniform in quality, clean, smooth, sound, and free from foreign material, blowholes, slivers, laps, cracks, seams, scale, burrs, buckles, damaged ends and edges, and other defects which will detrimentally affect the serviceability of the couplings.

3.10.1 Castings. Castings shall be sound and free from patching, misplaced coring, warping, and other defects which might render the casting unsound for its intended use.

3.10.2 Machine work. Tolerances and gauges for metal fits shall conform to the limitations specified herein and on the applicable figures.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, and unless disapproved by the Government, the contractor's own or any other facilities suitable for the performance of the inspection requirements specified herein, may be used. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the document shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known

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defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for ensuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 First article examination. The first article shall be examined as specified in 4.5.1, as applicable. Presence of one or more defect shall be cause for rejection.

4.3.2 First article tests. The first article shall be tested as specified in 4.5.2.1 and 4.5.2.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.4.2 Examination.

4.4.2.1 Individual. Each coupling shall be examined for the critical and major defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2.2 Samples. Samples selected in accordance with 4.4.1 shall be examined for the major defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.1 and 4.5.2.2. Failure of any test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The coupling assembly shall be examined as specified herein for the following defects:

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Critical

1. Holes in casting (see 3.3.3.1).

Major

101. Dimensions not as specified (see 3.1).
102. Material not as specified (see 3.3).
103. Materials are not resistant to corrosion or deterioration, or treated to be made resistant to corrosion or deterioration for the applicable storage and operating environment as specified (see 3.3.1).
104. Contractor does not have documentation available for identification of materials, material finishes, or treatments (see 3.3.2).
105. Finish not as specified (see 3.4).
106. Corrugations not as specified (see 3.5).
107. Threads not as specified (see 3.6).
108. Incorrect or missing parts (see 3.7).
109. Identification marking missing, incomplete, or illegible (see 3.9).
110. Workmanship not as specified (see 3.10).

4.5.2 Test.

4.5.2.1 Hydrostatic test. Subject the complete coupling assembly to a hydrostatic pressure of 400 psig for not less than 30 seconds. Examine the coupling assembly for leakage, rupture, and yielding. Leakage, rupture, or yielding shall constitute failure of this test.

4.5.2.2 Spectro-chemical analysis. Not less than a 2-ounce metal specimen shall be taken from each lot of couplings for analysis in accordance with ASTM E 172. Failure to meet the requirements of 3.3.4 shall constitute failure of this test.

4.6 Inspection of packaging. The preservation, packing and marking shall be examined and tested in accordance with the quality assurance provisions of MIL-H-775.

5. PACKAGING

5.1 Preservation, packing and marking. The levels of preservation, packing and marking shall be in accordance with MIL-H-775, or as specified by the contracting activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The couplings are intended for use in joining hose in steam systems, under 250 psig.

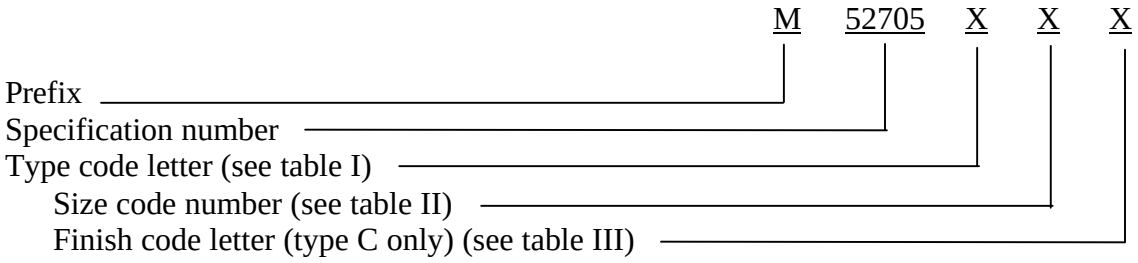
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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type and size coupling required (see 1.2).
c. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required for inspection and approval (see 3.2 and 4.3).
e. When type C coupling components, except washers and soft metal contact surface inserts, shall not be zinc plated (see 3.4.1).
f. Level of preservation and packing required (see 5.1).
g. Part or identifying number (PIN).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Part or identifying number (PIN). The part or identifying number (PIN) of coupling covered by this specification shall be designed in the following form (see 6.2):



6.4.1 Type coupling designation. The coupling type is identified by a single alphabetic character (see table I).

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TABLE I. Type code letter.

Code number	Type
A	A
B	B
C	C

Type A - Expansion ring coupling with slotted swivel.
Type B - Ribbed shank coupling with hexagonal swivel.
Type C - Ribbed shank coupling with wing nut swivel.

6.4.2 Size. The size of the coupling is identified by a single digit number (see table II).

TABLE II Nominal size code number.

Code number	Nominal size (in inches)
1	1/2 (.50)
2	3/4 (.75)
3	1 (1.00)
4	1-1/4 (1.25)
5	1-1/2 (1.50)
6	2 (2.00)
7	2-1/2 (2.50)

6.4.3 Finish. Finish for the type C coupling shall be identified by a single alphabetic letter (see table III).

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TABLE III. Finish code letter.

Code letter	Finish
A	Without plating
B	With zinc plating

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.6 Subject term (key word) listing.

Coupling, Expansion Ring
Fittings, Hose, Steam
Coupling, Ribbed Shank
Fittings, Steam, Hose

Custodians:
Army – ME
Navy – YDl
Air Force – 99

Preparing activity:
Army – ME

Project 4730-0149

Review activities:
Army – GL
Air Force – 82
DLA – CS

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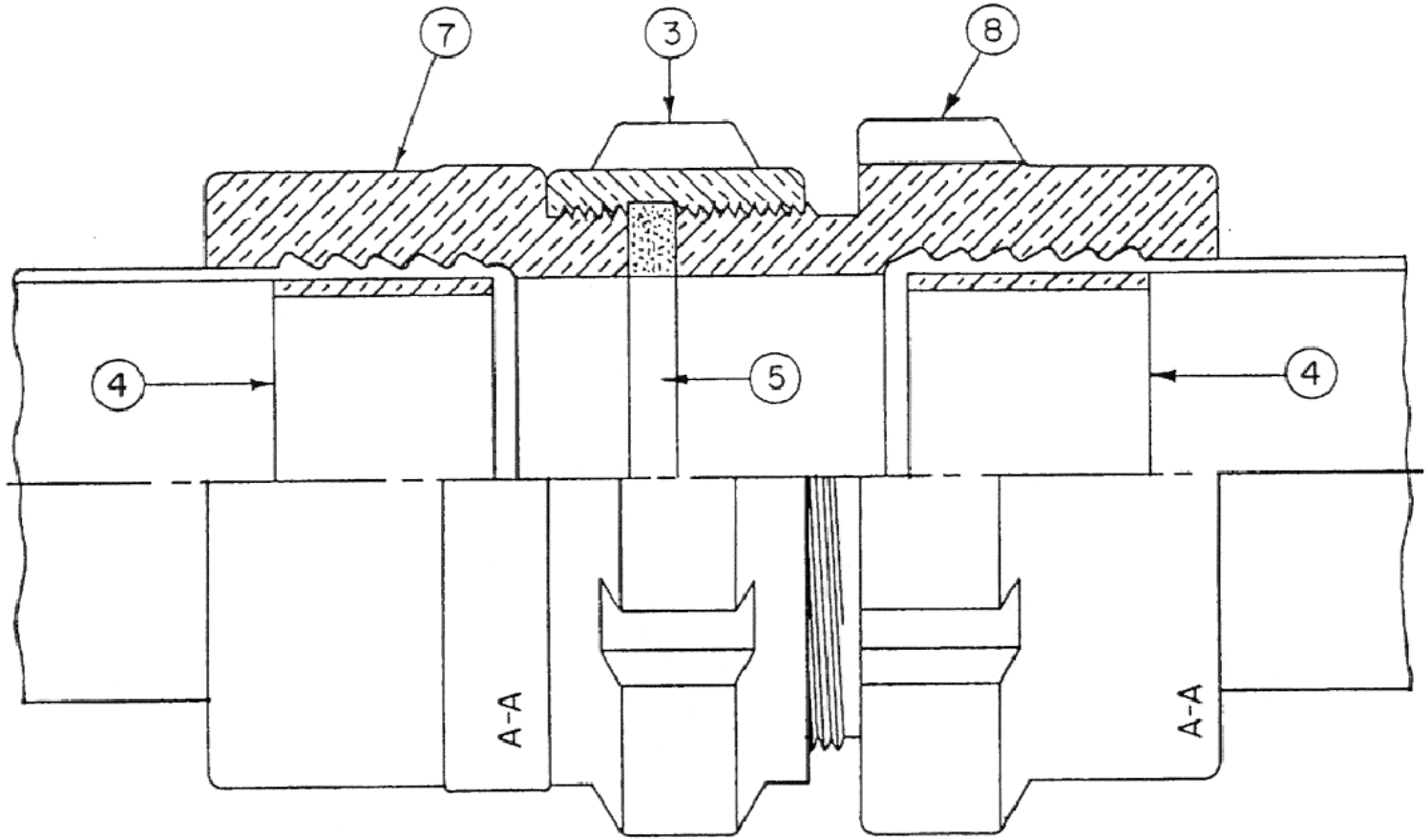
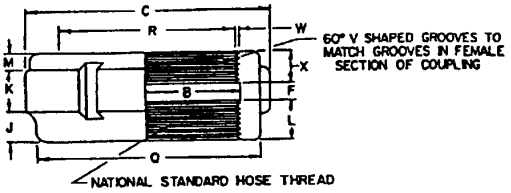
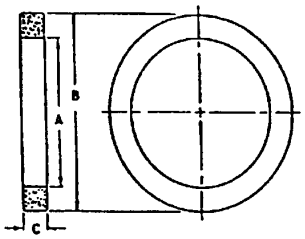


Figure 1. COUPLING ASSEMBLY-TYPE A.

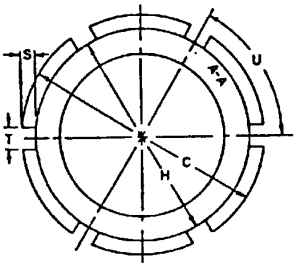
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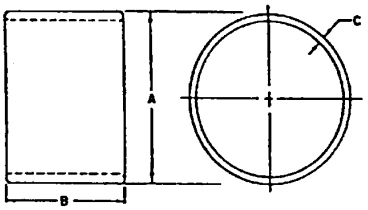




RUBBER WASHERS
PART 5

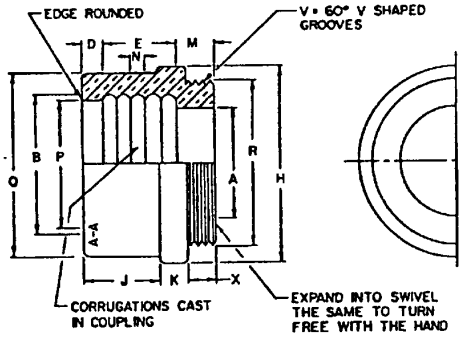


SWIVEL PART 3



EXPANDING RING
PART 4
EDGES WELL ROUNDED AND FREE FROM BURRS

COUPLING FEMALE SECTION
PART 7



COUPLING MALE SECTION
PART 8

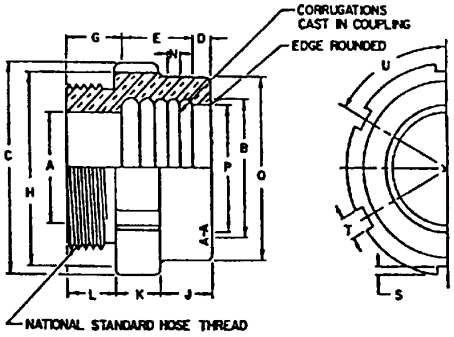


TABLE OF PARTS FOR (3/4 INCH) STEAM HOSE COUPLING

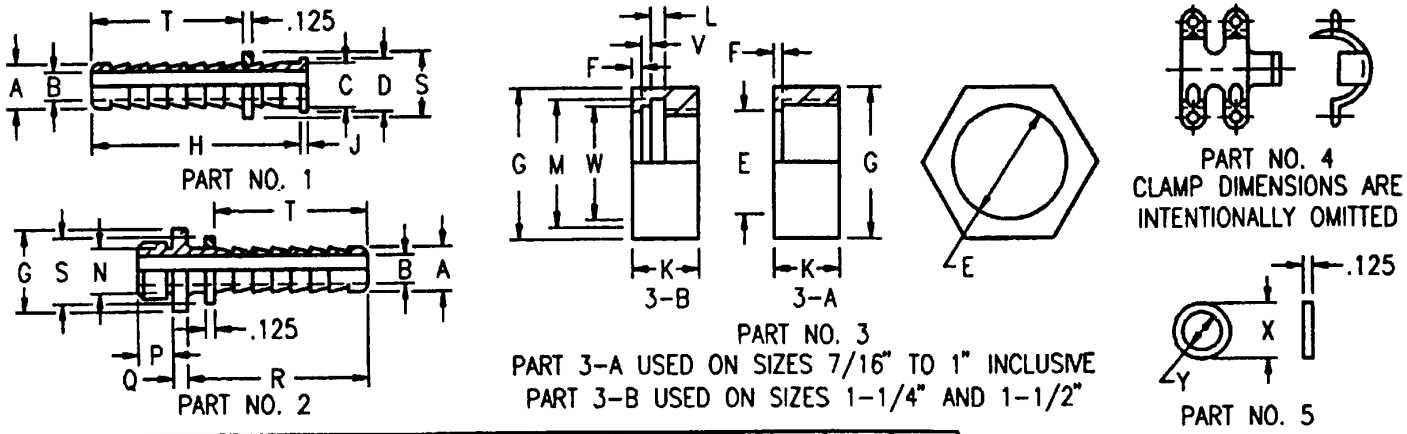
Part No.	A	B	C	D	E	F	G	N	J	K	L
2 and 8	.875	-------	2.250	.188	1.500	----	.938	2.000	1.250	.625	.750
3	-----	1.438	2.250	-----	-----	.250	-----	2.000	.250	.825	.900
4	.750	1.250	.04/.05	-----	-----	----	-----	----	----	----	----
5	.812	1.406	.188 (.125 MIN)	-----	-----	----	-----	----	----	----	----

Part No.	M	N	P	Q	R	S	T	U	V	W	X
2 and 8	.500	.188	1.500	1.938	1.312	.125	.250	60°	4	.062	.312
3	.188	-------	-----	1.938	1.312	.125	.250	60°	4	.062	.312

FIGURE 2. Steam hose coupling Type A.

X-2002A

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SIZE (NOM)	A	B	C	D	E	F	G	H	J	K	L	M
1/2 (.50)	.562	.375	.594	.719	.609	.094	1.094	2.750	.094	.656	–	–
3/4 (.75)	.812	.562	.828	.938	.844	.125	1.312	2.938	.094	.688	–	–
1 (1.00)	1.047	.812	1.062	1.172	1.078	.125	1.656	3.000	.094	.812	–	–
1-1/4 (1.25)	1.312	1.062	1.344	1.484	1.359	.125	2.000	3.438	.125	.875	.188	1.688
1-1/2 (1.50)	1.562	1.312	1.594	1.734	1.609	.125	2.219	3.688	.125	.875	.188	1.938

SIZE (NOM)	N	P	Q	R	S	T	U	V	W	X	Y
1/2 (.50)	RELIEVE AT LEAST TO BOTTOM OF THREAD	.438	.188	2.375	.875	2.000	.500	–	–	.734	.500
3/4 (.75)		.500	.188	2.812	1.312	2.250	.750	–	–	.938	.688
1 (1.00)		.562	.250	3.000	1.594	2.250	1.000	–	–	1.188	.938
1-1/4 (1.25)		.625	.250	3.125	1.875	2.500	1.250	.125	1.500	1.625	1.250
1-1/2 (1.50)		.625	.250	3.562	2.344	2.750	1.500	.125	1.750	1.875	1.500

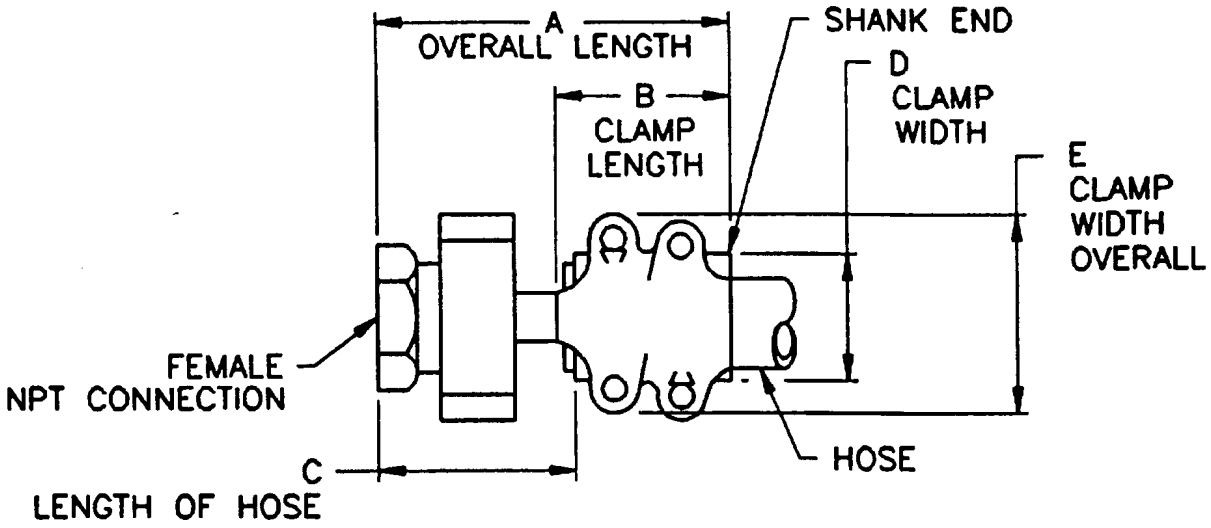
NOTES:

1. "U" DENOTES NOMINAL SIZE OF HOSE THREADS.
2. ROUGH CASTING TOLERANCES:
   INSIDE $^{+.000}_{-.016}$ OUTSIDE $^{+.016}_{-.000}$
   FINISH: ±0.005

FIGURE 3. Steam hose couplings–Type B.

X-2003B

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PART 9

NPT	HOSE SIZE	A	B	C	D	E	SHANK DIA IN HOSE	SHANK LG IN HOSE	WT IN LBS OF COUPLING
1/2	1/2	3.687	1.813	2.063	1.312	2.125	.531	1.625	1.0
3/4	3/4	4.875	2.750	2.500	1.813	2.750	.781	2.375	2.8
1	1	5.250	3.125	2.500	2.312	3.375	1.031	2.625	3.6
1-1/4	1-1/4	6.500	3.750	3.375	4.000	5.000	1.281	3.250	6.7
1-1/2	1-1/2	6.625	3.875	3.375	4.125	5.125	1.563	3.375	6.9
2	2	7.250	4.250	3.750	5.063	6.063	2.031	3.938	11.0
2-1/2	2-1/2	8.500	5.000	4.250	6.375	7.250	2.563	4.500	18.7

FIGURE 4. Steam hose couplings-Type C.

X-2082B

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